3.8.2 Gene expression is controlled by a number of features (A-level only)

3.8.2.1 Most of a cell's DNA is not translated (A-level only)

Content	Opportunities for skills development
Totipotent cells can divide and produce any type of body cell. During development, totipotent cells translate only part of their DNA, resulting in cell specialisation. Totipotent cells occur only for a limited time in early mammalian embryos. Pluripotent cells are found in embryos; multipotent and unipotent cells are found in mature mammals and can divide to form a limited number of different cell types. • Pluripotent stem cells can divide in unlimited numbers and can be used in treating human disorders. • Unipotent cells, exemplified by the formation of cardiomyocytes. • Induced pluripotent stem cells (iPS cells) can be produced from adult somatic cells using appropriate protein transcription factors. **Students should be able to** evaluate the use of stem cells in treating human disorders.	**AT i** Students could produce tissue cultures of explants of cauliflower (*Brassica oleracea*).

3.8.2.2 Regulation of transcription and translation (A-level only)

Content	Opportunities for skills development
In eukaryotes, transcription of target genes can be stimulated or inhibited when specific transcriptional factors move from the cytoplasm into the nucleus. The role of the steroid hormone, oestrogen, in initiating transcription. Epigenetic control of gene expression in eukaryotes. Epigenetics involves heritable changes in gene function, without changes to the base sequence of DNA. These changes are caused by changes in the environment that inhibit transcription by: • increased methylation of the DNA or • decreased acetylation of associated histones. The relevance of epigenetics on the development and treatment of disease, especially cancer. In eukaryotes and some prokaryotes, translation of the mRNA produced from target genes can be inhibited by RNA interference (RNAi). **Students should be able to**: • interpret data provided from investigations into gene expression • evaluate appropriate data for the relative influences of genetic and environmental factors on phenotype.	

3.8.2.3 Gene expression and cancer (A-level only)

Content	Opportunities for skills development
The main characteristics of benign and malignant tumours. The role of the following in the development of tumours: • tumour suppressor genes and oncogenes • abnormal methylation of tumour suppressor genes and oncogenes • increased oestrogen concentrations in the development of some breast cancers. **Students should be able to:** • evaluate evidence showing correlations between genetic and environmental factors and various forms of cancer • interpret information relating to the way in which an understanding of the roles of oncogenes and tumour suppressor genes could be used in the prevention, treatment and cure of cancer.	